```
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import (train_test_split, GridSearchCV)
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier
from sklearn.metrics import (accuracy_score, precision_score,recall_score, f1_score)
# 1. Load dataset
data = pd.read_csv("C:\\Users\\Avishkar\\OneDrive\\Desktop\\AIML7\\wdbc.csv")
X = data.drop(columns=["target"])
y = data.target
# 2. Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,random_state=42,
stratify=y)
# 3. Define models and parameter grids
models = {
    "SVM": (Pipeline([("scaler", StandardScaler()),("model", SVC())]),
        {
            "model__C": [0.1, 1, 10],
            "model__kernel": ["linear", "rbf"]
        }
    ),
    "Random Forest": (
        RandomForestClassifier(random_state=42),
```

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        {
            "n_estimators": [50, 100],
            "max_depth": [None, 5, 10]
        }
    ),
    "AdaBoost": (
        AdaBoostClassifier(random_state=42),
        {
            "n_estimators": [50, 100],
            "learning_rate": [0.5, 1.0]
        }
    )
}
results = []
# 4. Train, evaluate and optimize each model
for name, (model, params) in models.items():
    # Before optimization
    model.fit(X_train, y_train)
    pred_before = model.predict(X_test)
    before_accuracy = accuracy_score(y_test, pred_before)
    before_f1 = f1_score(y_test, pred_before)
    # Hyperparameter tuning
    grid = GridSearchCV(
        model,
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    params,
    cv=5,
    scoring="accuracy",
    n_jobs=-1
)
grid.fit(X_train, y_train)
# After optimization
best_model = grid.best_estimator_
pred_after = best_model.predict(X_test)
results.append({
    "Model": name,
    "Accuracy Before (%)": round(
        before_accuracy * 100, 2
    ),
    "F1 Before": round(before_f1, 4),
    "Accuracy After (%)": round(
        accuracy_score(y_test, pred_after) * 100, 2
    ),
    "Precision After": round(
        precision_score(y_test, pred_after), 4
    ),
    "Recall After": round(
        recall_score(y_test, pred_after), 4
    ),
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        "F1 After": round(
            f1_score(y_test, pred_after), 4
        ),
        "Best Parameters": str(grid.best_params_)
    })

# 5. Display results
df = pd.DataFrame(results)
print(df.to_string(index=False))
```

Output:

(venv) PS C:\Users\Avishkar\OneDrive\Desktop\AIML7> python 6.py

Model Accuracy Before (%) F1 Before Accuracy After (%) Precision After Recall AfterF1 After Best Parameters

SVM 98.25 0.9861 98.25 0.9861 0.9861 0.9861 {'model__C': 0.1, 'model__kernel': 'linear'}

Random Forest 95.61 0.9655 95.61 0.9589 0.9722 0.9655 {'max_depth': None, 'n_estimators': 100}

AdaBoost 95.61 0.9660 95.61 0.9467 0.9861 0.9660 {'learning_rate': 0.5, 'n_estimators': 100}